IEEE smart village®

Power a Village, Empower Community

Portfolio of Project Briefs - 2022

IEEE Smart Village's unique approach support the world's energy-impoverished communities by providing a comprehensive solution of renewable energy, community-based education, and entrepreneurial opportunities.

The ISV team needs your help. Volunteers enable power, increase educational opportunities, and support entrepreneurs in the remotest areas of the world.

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Introduction and Purpose

The Institute of Electrical and Electronics Engineers (IEEE) is the leading professional organization for the advancement of technology. With more than 420,000 members in more than 160 countries, it is the world's largest technical professional society, supported by the IEEE Foundation[1] and a global network of volunteers and philanthropic donors

IEEE Smart Village (ISV) is a unique venture dedicated to empowering marginalized communities through access to reliable electricity, broad-based education, women's empowerment, and market-based business development. ISV works to foster *sustainable* development to improve lives in impoverished villages worldwide.

There are three key messages that we hope you understand after your review:

IEEE Smart Village has crafted a studied, effective, systematic, and unique approach to sustainable development.

> ***Our core operations and oversight functions are almost exclusively fueled by volunteers. We work with remarkable, committed, and inventive people on the ground in remote parts of the world.***

The purpose of this booklet is to acquaint you with ISV, our works, and methods to combat global poverty, our on-the-ground Non-Governmental Organizations (NGOs) who execute the projects, and how you can support our crusade.

Local community members work to install an ISV-funded micro-utility in Cameroon.

[1] The IEEE Foundation, a tax-exempt 501(c)(3) organization in the United States, fulfills its purpose by soliciting and managing donations, recognizing the generosity of our donors, supporting high impact IEEE programs, and awarding grants to IEEE grassroots projects of strategic importance. IEEE Foundation serves as a steward of donations that improve the human condition, empower the next generation of engineers and scientists, educate and raise awareness, energize and recognize innovation, and preserve the history of technology. With donor support, the IEEE Foundation strives to be a leader in transforming lives through the power of technology and education.

Among the world’s most impoverished, nearly 1 billion people still lack access to electricity. Over 10 years ago, an organization that was to become IEEE Smart Village began to pursue a studied approach to addressing this population. Our goal was to not merely provide aid, but to develop economically self-sustaining communities. Continuously refined, what we have developed is a comprehensive and systematic approach to development. Working to “bring the world to the village”, the approach leverages advanced technology for energy, communications, water, and sanitation, and absorbs and shares lessons learned and best practices in economic and educational development. A primary goal of the approach is to render the local development sufficiently comprehensive that people, particularly young people, will determine that the local environment affords more opportunity than that offered in an urban area and remain in the village.

We “bring the world to the village”

The approach is encapsulated by ISV’s Three Pillars of Sustainable Development: Energy, Education, and Entrepreneurship, which all ISV projects must address. While the ISV pillars directly correspond to United Nations Sustainable Development Goals (SDGs) 4, 7 and 8 (see figure below), in practice their ripple effects touch almost all other UN goals.

ISV’s Three Pillars of Sustainable Development directly correspond with UN SDGs 4, 7 and 8, and nurture several others.

ISV target communities are either not served or are poorly served by a national electrical grid. Therefore, our Energy Pillar stems from providing solar-based electricity. For an impoverished community, electricity is fundamentally transformational and a catalyst for positive change. For example, electricity can offer clean, reliable lighting - ending reliance on filthy and dangerous kerosene; improved access to clean water and better sanitation; and underpin large boosts to productivity.

Sources of productivity boosts include extended hours for work and/or study; irrigation; the availability of labor-saving power tools; easy and immediate phone and device charging; and even access to the Internet.

Electricity is *Transformational*. In India, new light at a 12th century monastery (left); the first satellite connection for education (right).

Recognizing the importance of electricity to the mission, ISV and the IEEE designed and packaged a small (~1 KW), portable solar power station. Called "SunBlazer", it was highly cost-effective and could be deployed – or redeployed - with minimal skill. It is an example of how ISV overcomes everyday obstacles to development. Continually updated as technology advances, SunBlazers are still part of the ISV toolkit as a low density, rapidly-deployable energy solution. As costs in solar power equipment have fallen significantly in recent years, ISV's on-the-ground NGO entrepreneurs increasing realize the Energy Pillar with AC/DC micro-grids. However, the Portable Battery Kits (PBKs) developed for the SunBlazer remain a common part of the energy solution and our NGOs' local energy businesses.

A consideration with mini or micro-grids is whether they can be integrated with the national grid once the latter is extended to area. Investing to develop a micro-grid can be less attractive if there is uncertainty that the investment can be stranded if the grid arrives. Anticipating this – and protecting its own investment and that of its donors – ISV's systematic approach yet again comes into play. As many of the engineers who work for national power utilities are also IEEE members, ISV works with these members to plan for the ultimate integration of the local micro-girds with the national grid.

"...transformational and a catalyst for positive change"

With solar-derived energy in place, ISV's other Pillars of Sustainable Development, Education and Entrepreneurship, come to the fore. Education endeavors to serve children (pre-K to 12) with digital classrooms supported by both Internet and private Intranets. Education must promote gender equality. For adult education and training, ISV took a studied approach and determined that an SECI-based (socialization, externalization, combination, and internalization) approach, was the right one for the on-the-ground situation.

Heavily reliant on case studies and increasingly-independent problem solving, the SECI approach is well suited for adults with immediate needs to apply the newly-acquired knowledge, which is the operative situation in the villages. Vocational training is delivered by a combination of local subject matter experts and Internet based online training.

As the micro-grid will always need skilled, local hands-on support, it is an inherent vehicle to train and employ village members. These classes typically cover the basic principles of electricity needed for solar power distribution, maintenance of the microgrid, in-home wiring, and installation and operation of community power distribution. Other vocational topics depend on the village's specific situation. These have included brick-making and construction, beekeeping, beadwork, computer skills, customized agricultural training, financial management, food dehydration and processing, retail operation, and dealing with the water and sanitation systems. This comprehensive approach leads to community ownership and support of the ISV sponsored business. Within the context of ISV's systematic approach, training and specific village case studies become part of a cloud-based knowledge base that all ISV projects can share.

"...attack the roots of poverty"

Education for entrepreneurial-driven economic development can also include higher learning. Many leaders of our projects have pursued the fully accredited Master of Development program of the IEEE Global Classroom in Denver, CO. This program is delivered to villages worldwide for 1% of the cost of that at a top US university.

Overall, ISV's comprehensive and systematic approach to development promotes self-sustaining communities. Building upon the transformational nature of electricity and harnessing technology, the new associated educational and economic opportunities attack the root causes of poverty.

IEEE Smart Village Projects in Action

As suggested above, activities on the ground are always affected via an NGO with which ISV executes a project agreement. This not only harnesses local knowledge and conditions, but keeps costs low, thereby stretching development funding. Given their criticality to a project's success, all NGO candidates must submit a proposal outlining their project's approach to comprehensive development that ultimately can scale to reach up to one million people. The ISV evaluation team works with the NGO to refine its plans – always drawing upon best practices and lessons learned from other development efforts. In accepting initial funding, NGO entrepreneurs commit to growing their company by reinvesting in their local economy

As one might image, ISV's NGO project leaders are extraordinary people. Highly committed, they take their program responsibilities to ISV to heart and, in action, regularly demonstrate their inventiveness and willingness to tackle problems. Many of these leaders have degrees and could work in Europe or North America but choose to work in their home countries. Some of them have even returned to their native countries from abroad.

Project agreements are highly structured with funds disbursed in tranches tied to measures of program success. Reports are submitted quarterly to the ISV program manager, who presents overall progress and status to the ISV leadership team.

We note that outside of the ISV program manager, the whole of the ISV support team is comprised almost exclusively of volunteers, nearly all of whom are seasoned professionals. As ISV operates within the IEEE Foundation, the latter provides limited staff support to ISV, notably including legal assistance and administrative support.

The combination of dedicated, locally operating NGOs; tight project oversight and control, a nearly all-volunteer ISV team, and a well-honed, systematic approach to development yields measurable impact and overall, an outstanding return on funding.

Options for Making a Donation and a Difference

We have discussed ISV's well-considered, systematic approach to sustainable development and how successful our approach has been. We have related how almost all ISV "staff members" are volunteers – and the dedication of the NGO's making it happen on the ground. Now, let's look at how you can be part of making this happen!

How you can support the efforts of IEEE Smart Village:

1. **Contribute without making a designation to a specific ISV project:**

   Unrestricted contributions are the 'life-blood' of any charitable organization and provide flexibility of supporting a broad-range of ISV budgeted objectives. To learn how varying donation levels provide impact, visit https://www.ieeefoundation.org/SmartVillage_donation

   If you are considering a donation of US$10,000 or more and want to have a confidential discussion regarding giving opportunities and benefits, then please email us at donate@ieee.org or call +1-732-562-3915.

2. **Gifts of US$10,000 or more designated for one of the projects discussed below:**

   Designated Gifts need special handling to coordinate with IEEE Foundation. Contact Michael Deering, Senior Development Officer, m.deering@ieee.org, who will coordinate with IEEE Smart Village Sr. Program Manager Mike Wilson, m.m.wilson@ieee.org, to direct your funds, introduce you to Project Leaders, provide more detailed information that includes quarterly reports, and to forward requests for individual or small group site visits.

3. **Donate Time and Passion:**

   Because ISV is run almost solely by volunteers, we need a pool of volunteers from which to draw for specific tasks. Not all our volunteers have years of experience, but all of them understand the impact of our work and feel a personal commitment to help. We need help across the board, including Development, Marketing, Finance, Education, Technology, Operations and Engagement Committees

Here's how to start: http://ieee-smart-village.org/get-involved/volunteer/

Information on the IEEE Foundation and donations: ttps://www.ieeefoundation.org/disclosures-for-charitable-donation



Kenya, Turkana - The Natoot Farm Upscale Project

Developer: Full Gospel Church of Kenya, Turkana **Project Agreement # PSA 058 - 21** **Date: Jan 25th 2022**

Project Total Cost: US$ 24,000.00 **ISV Seed Funding: US$ 24,000.00** **Self-Funded & Pro Bono: US$ 14,000.00**

PURPOSE

The Natoot Farm is a pillar of hope for the many families that farm for sustenance and commerce in Lodwar, Turkana Central, one of the poorest counties in Northern Kenya. This project has improved food security, food affordability, food quality and socio-economic development. The Natoot Farm is situated in Lodwar in Turkana Central. Its GPS coordinates are 3.120891,35.650710. The farm is supported by Bright Hope International in partnership with the Full Gospel Churches of Kenya. Natoot Farm hosts over 70 farmers and is subdivided into 11 blocks with 7 to 8 farmers per block. As a model sustainable agriculture zone, the farm has attracted diverse individuals including college and university students whose contributions have greatly improved the performance of the farm.

OBJECTIVE

1. Provide increased Solar PV capacity to ensure sufficient water for crop irrigation and community needs.
2. Provide additional capacity for water storage.
3. Enhance security through solar powered security lights.
4. Improve the productivity of the farm and increase the amount of food production for commerce.
5. Improve community livelihoods and promote food security.

BACKGROUND

Turkana people are nomadic pastoralists who have depended upon cattle, donkeys, camels, and goats as their primary source of food and wealth. Sadly, Turkana is one of the poorest counties in Kenya despite high value unexploited resources. Turkana's climate is hot and arid. The Natoot Farm is changing the narrative by introducing fresh produce for families and markets. Bright Hope works together with their local partner church, Full Gospel Churches of Kenya and Turkana County to implement these partnerships in the communities through Core Groups. The Core Group and the community members work together to identify solutions to their most pressing challenges.

LEVERAGE AND SCALABILITY PLANS

ISV has teamed up with Bright Hope International, together with religious leaders and community members, to execute this project. In line with ISV's mission, Bright Hope shares the goal of developing sustainable projects in the community and understands that time and resources spent only on relief efforts may lead to an unhealthy dependency. Therefore, sustainability is embedded in the project's goals. The community members learn sustainable farming methods and techniques via a modern farm powered by solar energy. Each farmer is assigned land, individual knowledge and skills are utilized allowing for a competitive yet cooperative approach to farming with the aim to provide better food and products for market.

TEAM

BrightHope International, Turkana County and the IEEE are working together to ensure project success. The following people are key in the implementation of this project.

- Bishop Francis Namuya provides leadership fostering inspiration and responsibility within the community. The model of integration through church creates relevance to the villagers and the land in use belongs to the church.
- David Namii Napereng is the Food-Water Core group Team Leader in the congregation.
- Julius Oluoch is the Food-Water Core Group reporter and provides updates on the Core Group activities.

- Victor Juma is the Community Project Coordinator leading the farm and produce sales.
- Joseph Mulongo is the Farm Assistant helping farmers with daily operations.
- Paul Kanja is an IEEE Volunteer and energy consultant with GIZ helping to guide on the technicalities of the farm.
- Mercy Chelangat, IEEE Coordinator and liaison with Bright Hope provides additional updates and information.
- Dotun Modupe, the Bright Hope Country Manager Kenya.

IMPACT AND TRANSFORMATION

Expected Output: Families and Farmers with the ability to produce better crop yields ensures Self-Sustainability.
Expected Outcomes: 12,000 individuals or 2,000 families will have access to clean water. Approximately 200 smallholder's families will depend on subsistence farming for their livelihood.
Expected Impact: Available water, improved food security, the gains from marketing of additional produce to cover other needs like school fees and medical care.
Alignment with ISV 3 pillars: The solar power provided will give farmers access to a dependable source of water for irrigation. Increasing the income of farmers to cover the needs of family and the community contributes to the running and sustainability of the project.

TIMELINE

The project will take approximately one month to mobilize, implement and launch for utilization. The project timeline will involve; site assessment and technical design (week 1), delivery of materials and inspection (week 2), construction and assembly work, structural work and electrical work (week 3 & partially week 4), *project commissioning (week 4).

AT A GLANCE

Project Locations:	Lodwar, Turkana, Kenya
GPS	Latitude: 3.120891 Longitude:35.650710
IEEE Funding	US$ 24,000.00
Payment Tranches	Tranche 1: 60%, 2: 25%, 3: 15%, 4: 10%
# local IEEE members involved	3
Number of Beneficiaries (Phase 1)	200 smallholder families will benefit from the increased farm productivity.
Number of Beneficiaries (At scale)	20,000 community members with access to water, improved nutrition, boosted livelihoods.
Technologies to be deployed	Solar micro-grid to power drip irrigation
IEEE Review and Reporting Period	After project completion and quarterly every year
Sustainability/ Scalability	Project is for for technical advancement, surveys and other opportunities
Project Contact	Dotun Modupe - dotun.modupe@brighthope.org
Organization information	Bright Hope International and Full Gospel Church of Kenya, Turkana County

Zambia, Ndola - KSV Women's Gemstone Training

Developer: Kuumba Smart Vision Zambia - KSV Zed **Project Agreement # PSA 057 - 22 Date: Feb 18, 2022**

Project Total Cost: US$ 50,000.00 **ISV Seed Funding: US$ 24,475.00** **Self-Funded & Pro Bono: US$ 25,575.00**

PURPOSE

Primary purpose is education and training in business skills and in the processing of gemstones, equipping women to set up businesses in their villages around Ndola, Zambia. The lead organization, Kuumba Smart Vision Zambia (KSV-Zed), has signed an MOU with Gemstone Processing and Lapidary Training Centre (GPLTC), and Kariba Minerals (KML) in order to implement this project. KSV-Zed will train women from villages surrounding the mining towns, providing a path to succeed through business training and access to lapidary equipment. A solar energy system will be installed in the education center.

OBJECTIVE

1. Upgrade the program at GPLTC.
2. Train 40 women/year and improve curriculum
3. Build parts of the upstream and downstream ecosystem to enable women's gem-cutting enterprises.
4. Foster an understanding of the gemstones in Zambia and how the people can add value to stones to create jobs.
5. Upgrade lapidary equipment through electrical motor replacement
6. Install a renewable solar system to provide consistent and reliable service.

BACKGROUND

The broader community is the gemstone industry in Zambia, who recognize the need to add jobs for women and add value to the stones, rather than exporting the raw material. For this reason this has been a hand in hand partnership with those familiar with the industry and the struggles it faces. The specific community for this test is Ndola and the surrounding area, because of the proximity to the school and local artisan (small business owner) mines. The project team includes the school's principal, lead instructor, a woman who is the Vice President of the Federation of Small-Scale Mining Association and previously the president the Zambian in Women Mining Association. The MoU was approved by the Education Ministry of Zambia have granted the right to work with them and their facilities.

LEVERAGE AND SCALABILITY PLANS

KSV-Zed, a nonprofit NGO in the Republic of Zambia has been formed for this program. The partnerships and MOU are solidified and the donors for the project are secured.

The gemstone school has several indicators they use for determining success. Those will be used, as well as following up with surveys from recent graduates. The risks are very low on this development project in that GPLTC is an established school, adding capacity allow more women to attend this trade school.

The first phase of the project will be to engage women students and educate them to run successful businesses. The second phase is to work with the school to create a one year curriculum to allow a fast track for students.

The trade school has the capacity to train additional students, with success it can be easily expanded and funded from gemstone revenues.

KSV-Zed has a dedicated employee that will lead the project and coordinate with the local community. The cost of the employee's salary is not part of this proposal and is paid from other sources.

KSV-Zed is operating as a nonprofit and will use the revenues to sustain and grow the business.

TEAM

- President of KSV-Zed - Kanekwa Kachinga: Editor for the Smart Village EmPower newsletter, she is also a seasoned international journalist.
- Advisor - Pauline Sialumba Mundia: President of Zambian in Women Mining, has served on various mining boards. She is V.P. of the Federation of Small-Scale Mining Association and Project Coordinator of the Zambian Gemstone online platform.
- GPLTC Principal: Victor Mulenga
- GPLTC Lead Instructor: Lameck J Thole

IMPACT AND TRANSFORMATION

Number of women initially expected to graduate annually is 40, with an expected retention in the industry of 75%. Current retention of graduates are on the order of 15%.

Projected number of precious stones cut and polished per woman per day each polishing about 20 karats per day with the number of gemstone cutters increasing by 5 people every year.

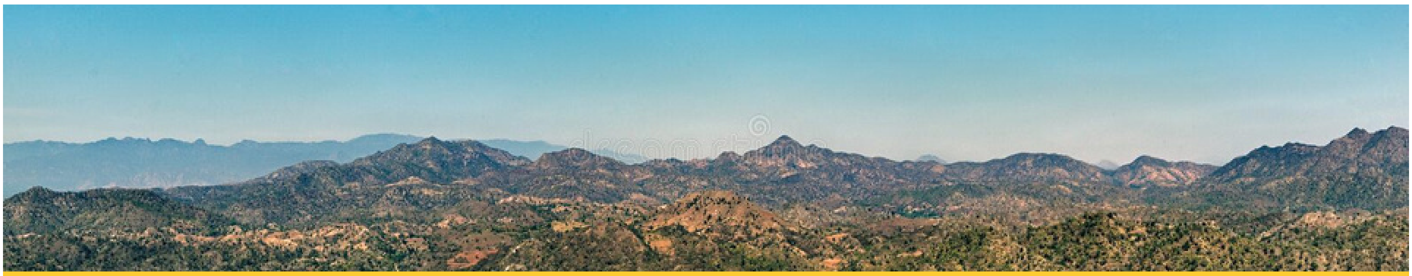
The impact for women trained in this trade and industry has the potential to improve individual livelihoods and the nation's industry and commerce. The synergy of education and training for women, their empowerment along with renewable solar energy is what makes this program likely to succeed.

TIMELINE

1.Upgrade old equipment Start: 01-May-2021 Completed: 30-June-2021 2. Install new equipment Start: 15-July-2021 Completed: 15-Aug-2021 (The installation date will depend on the equipment purchasing, shipping and clearing customs)
3. KSV-Zed training and eco-system Start: 01-Sep-2021 Completed: 15-Sep-2021. Student training Start: 01-Sep-2021
Ongoing 5. Solar System installation Start: 31-May-2022

AT A GLANCE

Project Locations:	Ndola, Copperbelt, Zambia
GPS	Latitude: -13.013750 Longitude: 28.654889
IEEE Funding	US$ 24,475.00
Payment Tranches	Tranche1: US$ 10,000, Tranche 2: US$ 2,500, Tranche 3: US$ 2500, Tranche 4: US$ 7500, Tranche 5: US$ 1,975
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	40 women in the community, local villages, and the broader gemstone industry.
Number of Beneficiaries (At scale)	Over the course of five years the first project will affect about 500 women. The program will grow.
Technologies to be deployed	Electricity, electrical equipment, hybrid solar, educational resources.
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	Project is for for technical advancement, surveys and other opportunities
Project Contact	Kanekwa Kachinga - kanekwa1@gmail.com
Organization information	Kuumba Smart Vision Zambia Plot 15 Poplar Avenue Plot 15 Poplar Avenue, Avondale, Lusaka, Zambia,



India, Aravalli - Digital Empowerment for Women and Adolescents

Developer: Magan Sangrahalaya Samiti **Project Agreement # PSA 061 - 22 Date: Q3 2022**

Project Total Cost: US$ 43,000.00 **ISV Seed Funding: US$ 20,500.00** **Self-Funded & Pro Bono: US$ 22,500.00**

PURPOSE

The project's focus will be to establish those conditions needed to educate and train young adolescent girls and women in the use of SMART technologies as part of the overall development initiatives of the Aravalli region. Digital technology amplifies and accelerates social, economic and environment development. This project targets a population of 21,000. Mahan Sangrahalaya Samiti (MSS) is a non government social service organization that began its work in 2003 with a commitment to empowering women in rural communities. MSS has supported several women's self-help groups in various micro enterprises.

OBJECTIVE

1. Create an environment for women entrepreneurs to connect and develop together.
2. Leverage women to participate in entrepreneurship using technology.
3. Encourage young women to consider careers in the field of information and communication technology.
4. Increase business opportunities for small scale industries in remote rural areas.

BACKGROUND

The intervention area, Aravalli rich in culture and environment, contains an abundance of resources.

Aravalli is inspired to transform and create business opportunities to address the current challenges it faces in employment, education, health, agriculture, nutrition, and other components required for a thriving community.

The consortium partners lead by MSS has cultivated an impeccable trust within the community enabling them encourage participation and contribution in project planning, implementation, and monitoring.

The project provides opportunities for participating communities allowing college students to develop digital skills. are Shamlaji College Gram Panchayat, a key project partner, shall provide infrastructure support and trainings working with Sarjan Foundation to prepare course modules and master trainers. Wheel Global Foundation in conjunction with Mission Samriddhi will provide technical and managerial support. MSS shall maintain all assets and keep ownership.

LEVERAGE AND SCALABILITY PLANS

The consortium of partners lead by MSS, has international and national partners representing several decades of experience in rural development, philanthropy through technologies, academics, and other aspects of development. The consortium will mobilize additional funding and other technical resources to scale up the project. A measurement framework has been developed identifying impact and key milestones. A baseline survey to document pre-intervention conditions will be conducted.

All the components of the project will generate income by charging a nominal fee to sustain the operations. The Digital lab fees collected from approximately 3,000 students are dedicated to maintaining equipment and to cover the cost of teachers. A rate card has been developed for the varied services. In addition to the services provided by the young adolescents, the project shall link participants with existing union government digital service centers to expand the utilization of technology by the villagers.

TEAM

- Dr. Vibha Gupta: Chairperson of the MSS, 40+ years of experience in rural enterprise development.
- Pradeep Kapur: Former Ambassador & Secretary at GOI (Government of India) and Executive Director of the Smart Village Development Fund, WGF USA.
- Dr. Ajay Patel: Principal of Shamlaji College with over a decade of leadership in college level education, including 7 Gram Panchayat Sarpanches.

IMPACT AND TRANSFORMATION

The Program combines enterprise and educational interventions to improve digital technology inclusivity for marginalized rural communities.

Program outcomes are as follows:
• 3,000 college students from Bhiloda Taluka have improved functional digital literacy through the Shamlaji College Digital Lab which provides digital literacy training to adolescents and young women utilizing Digital India and Skills India Programs.
• 7 Young Women New Age Digital Entrepreneur - Mobile Computer Digital Service Entrepreneurs are providing digital/online services to the local community in 7 Gram Panchayats, especially to the Sahkhi Mandals (Self-Help Groups).
• 21,000 Gram Panchayat (GP) members have expanded access to digital information, online services, and increased digital literacy.

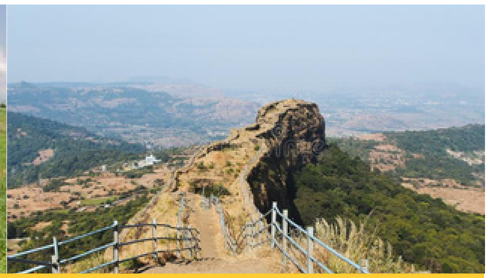
TIMELINE

- Project start date: 01 Jan 2022 and end date: 31 Dec 2024.
- A Digital Lab Established and Functional in Shamlaji College by Apr 2022.
- 7 YWNADEs trained and provide services by May 2022.
- 7 Digital functioning kiosks, access to services by Apr 2022.

AT A GLANCE

Project Locations:	Bhiloda Taluka
GPS	Latitude: 23.6878° N Longitude: 73.3874° E
IEEE Funding	US$ 20,500.00
Payment Tranches	Tranche 1: US$ 20,500
# local IEEE members involved	3
Number of Beneficiaries (Phase 1)	3007
Number of Beneficiaries (At scale)	21,000
Technologies to be deployed	Internet/Intranet; Computers, Digital Kiosks; Internet of Things; Digital Training Course (Online-Offline)
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Dr. Vibha Gupta - vibhawomentech@gmail.com
Organization information	Magan Sangrahalaya Samiti Kumarappa MArg, Wardha 442001, Maharashtra Kumarappa MArg, Wardha 442001, Maharashtra Wardha, Maharashtra 442001





India, Velhe - Transforming Agro Biomass Energy to Biofuel Pellets for Rural Household

Developer: Jnana Prabodhini **Project Agreement # PSA 062 - 22 Date: Q3 2022**

Project Total Cost: US$ 26,500.00 **ISV Seed Funding: US$ 25,000.00** **Self-Funded & Pro Bono: US$ 1,500.00**

PURPOSE

This project proposes a facility for the production of biomass pellets as a cooking fuel. Output will service 100 domestic consumers in the village of Velhe Khurd, Pune District, India. The pellet plant will be installed, supervised, and cooperatively operated. The pellets will be sold by women's Self-Help Group (SHG) in the village. The project includes education on the impact of producing and using agricultural waste as the source of biofuel for cooking. Surplus pellets produced in this pilot project will be sold to nearby villages creating awareness and developing nearby markets. The developer, Jnana Prabodhini (JP), means 'Awakener of Knowledge' in Sanskrit. Founded in 1962 with the motto "motivating intelligence for social change", its activities have expanded into multiple aspects of social work. The Rural Women Empowerment division of JP began working in Velhe Taluka in 1995. JP is now connected to 6,000 families and 225 SHGs.

OBJECTIVE

1. To establish a pellet production facility using local agricultural biomass.
2. Create employment utilizing local village manpower.
3. To provide a source of clean cooking fuel and to build a market and revenue stream for clean cooking fuel.
4. Reduce the dependence on firewood and the time intensive collection by the women.
5. Reduce the exposure to smoke inhalation and related health issues.

BACKGROUND

The village of Velhe Khurd has a population of 1,200. Income generation is primarily from agriculture indigenous farmers cultivate rice, ragi and other crops. The community actively participates in various educational and developmental programs implemented by JP. Along with the pellet production, the project provides extensive training on the energy potential of various raw materials, the basics of combustion, dangers of pollution and the consequential health and environmental impact of burning firewood.

The pelleting hardware will be secured from Gangotree Eco Technologies Pvt. Ltd. (GET). GET recently conducted trials providing sample pellets from locally sourced materials and provided clean cookstoves to field test the production capacity of the resource materials as well as test acceptance of the cookstove program. GET will train the Self-Help Group (SHG) in the operation, production, marketing, and sale of the cooking pellets. A portion of the net profits will be retained by the SHG for maintenance and repair of the pelleting machinery. A group formed with SHG community members along a JP representative will own the assets of the project. GET trained SHG members will run the operations overseen by JP.

LEVERAGE AND SCALABILITY PLANS

The locally available waste agricultural biomass is the source of raw material for production of the fuel pellets. Within the scope of this project, there are no legal restrictions in the use of agricultural waste for bio-fuel pellet production. Raw material cost will be minimal. Farms that currently have discarded agricultural waste have pledged to contribute to the program. Those farms currently burning their waste as a fuel may receive subsidies. Sales of pellets in nearby villages will drive revenue for the program. JP will act as an agent selling the pellet products in nearby villages through the local SHGs. Furthermore, we expect the introduction of low-cost pellets will promote use of pellets-based smokeless stoves, having a substantial positive health and environmental impact. JP's participation, mentorship and support commitment is for three years, assuring the project achieves operational and financial selfsustainability.

TEAM

- Suvarna Gokhale, MA, DBM, M. Phil. Member of Executive committee; active with JP Rural women department since 1995.
- Surekha Dighe Ex. Member of Grampanchayat of Velhe Khurd., supervises 25 SHGs across the Velhe region and is head of JP skill development, production, and marketing.
- Dr. Sunil Gokhale (Gangotree Eco Technologies Pvt. Ltd.), material scientist and pelleting expert.
- Amol Chiplunkar (AAC Energy Consultants) B.E. Production, M. Tech Energy, Certified Energy Auditor and technical consultant.

IMPACT AND TRANSFORMATION

Outputs: The fuel pellet facility will employ 3 people full time with a forecast production of 10 ton per month.

Outcomes: On an annual basis, 100 MT of agricultural waste will produce approximately 90 MT fuel pellets.

Target recipients of the pellets in the village will be split between financially secure families and the underprivileged section.

Across the 100 families engaged, expectation is that pelletized fuel will reduce consumption of wood burning by 70%.

Impact: Reduction in deforestation, reduction in home environment pollution, substantial reduction in rural women efforts to acquire fuel resources (dramatically increasing the time available for other gainful activities), households from adjacent villages will start using smokeless stoves.

TIMELINE

With project initiation in the 3rd quarter 2022 (subjected to release of funds) procurement of all hardware will be completed by then end of the 4th quarter of the year. Simultaneously awareness training, material collection, project infrastructure and power connection will be arranged. With delivery of the pelletizing equipment in the 4th quarter, commissioning is possible at the start of 2023. Distribution of stoves takes place concurrently as pellet production ramps up. Project is forecast to be fully operational by the end of the 2nd quarter of 2023.

AT A GLANCE

Project Locations:	Velhe Khurd , Pune District, Maharashtra State, India
GPS	Latitude: 18.28799 Longitude: 73.63379
IEEE Funding	US$ 24,475
Payment Tranches	Tranche 1: US$ 22,000, Tranche 2: US$ 3,000
# local IEEE members involved	TBA
Number of Beneficiaries (Phase 1)	100 Households (families) from one village
Number of Beneficiaries (At scale)	
Technologies to be deployed	Pelleting of waste Agro Biomass, Smokeless cooking stoves for pellets burning.
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	Project will be sustainable after 2 years
Project Contact	Suvarna Gokhale - suvarna.gokhale@jnanaprabodhini.org
Organization information	Jnana Prabodhini 510, Sadashiv Peth, Pune 411030 India Pune, Maharashtra 411030

India, Nagaland - Empowering the Naga Communities

Developer: Global Himalayan Expedition (GHE) **Project Agreement # PSA 060 - 22** **Date: Q2 2022**

Project Total Cost: US$ 39,035.00 **ISV Seed Funding: US$ 24,919.00** **Self-Funded & Pro-bono: US$ 14,116**

PURPOSE

Solar electrification of the Aboi Changle village in the state of Nagaland, India. A first of its kind for this village, the socioeconomic development accomplished at the local level will ensure the maintenance and functioning of the newly created infrastructure. The solar electrification of this village will impact the lives of 44 households and 377 people that reside in this remote village. Global Himalayan Expedition (GHE) will lead the project in partnership with the local community and government administration. GHE has previously partnered with IEEE Smart Village in the electrification of villages and monasteries in the Himalayas. This Project fits within the mission and vision of the IEEE and IEEE SV and directly furthers IEEE SV's goal of empowering rural communities in remote locations.

OBJECTIVE

1. Survey and mobilization of the community.
2. Transportation of the solar grid material to the site.
3. Creating a joint community bank account for the community for the project to enable the operations and maintenance of the solar grids.
3. Installation and commissioning of the solar grid equipment.
4. Encouraging the local community through training of the youth as solar technicians to operate and maintain the grid.
5. Empowering women to invest more time in basket making and embroidery and creating the market linkage.

BACKGROUND

Aboi Changle village is located close to the India-Myanmar border. The villagers come from the Konyak tribe, who are amongst the oldest tribes of the region.
The Village Head man, Mr. Phaiba and Village Chairman, Mr. Khole will be the main persons from the village, who will lead Aboi Changle Village.
Lead Organization Global Himalayan Expedition (GHE) is a social enterprise that has brought solar energy access to more than 140 villages across 5 regions of India.

GHE has already electrified its first village in Nagaland and is now working to scale up the operations in the region to develop the capacity in the communities. GHE will work alongside the community to create an impact model. It will also assist with the solar system design, setup and implementation in the village, while training the local communities.
The community will take advantage of the manpower and natural resources available such as bamboo and timber to prepare the structures to be used for the project.

LEVERAGE AND SCALABILITY PLANS

A village bank account will be opened in the name of Solar Energy and every household will contribute 100 Indian Rupees (INR) per month towards the maintenance and upkeep of the solar grids. Several youth will be identified in the village as part of the Village Management Committee (VMC) and they will be responsible for the monitoring the impact of solar electrification. They will also be responsible for conducting independent audits along with the government administration. The primary goal is to create a sustainable livelihood model through the use of solar energy.

Once the pilot is done, GHE will reach out to several national and international organisations to scale up the impact work which will also involve the following verticals
1. Solar Electrification of Villages
2. Solar Electrification of Medical Centres with Installation of Medical Equipment
3. Creating local entrepreneurs and piloting livelihood and agri based tools
4. Carbon Offset projects through Clean Cookstoves
5. Homestay tourism development

Due to a strong ground presence and having trained local entrepreneurs, GHE is able to mitigate the risks and also work in partnership with the administration for any necessary approvals.

TEAM

- Mr. Jaideep Bansal is the COO of GHE. He has over 10 years of experience in the solar energy and rural development sector and will coordinate the overall deployment of the project.
- Mr. John will be the primary point of local contact in Nagaland. He has been instrumental in getting GHE to Nagaland.
- The Village Head man, Mr. Phaiba and Village Chairman, Mr. Khole will be the main persons from the village who will support by engaging the community participation.

IMPACT AND TRANSFORMATION

- Total Households Electrified: 44
- Total Lives Impacted: 377
- Solar Capacity Installed: 7.5 kW
- Total CO2 Offset: 16.83 Tons Annually
- Improvement in Lux Levels: 500%
- Women Engaged in Handicrafts: 88
- Incremental hours of study for children: 3

GHE shall provide a detailed project documentation and quarterly reports to IEEE SV for the project period.

This will involve the documentation of technical grid design/specifications, hardware parameters, and any other needed reports.

Other documents such as business plans and operational methodology, impact assessment reports and community transformation data will be provided as well as digital images and videos.

TIMELINE

- Project Start – Jul. 2022
- Transport of Solar Material- Jul 2022
- Installation of Solar Grids- Sep 2022
- Project End – Oct 2022
- Training of Local Youth as Solar Engineers- Jul-Aug 2022
- Impact Assessment- Oct 2022

AT A GLANCE

Project Locations:	Aboi Changle, Mon/Nagaland, India
GPS	Latitude: 26.588451 Longitude: 94.907161
IEEE Funding	US$ 24,919.00
Payment Tranches	Tranche 1: US$ 19,935, Tranche 2: US$ 4,984
# local IEEE members involved	TBA
Number of Beneficiaries (Phase 1)	100 Households (families) from one village
Number of Beneficiaries (At scale)	
Technologies to be deployed	Pelleting of waste Agro Biomass, Smokeless cooking stoves for pellets burning.
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Jaideep Bansal - jaideep@ghe.co.in
Organization information	Global Himalayan Expedition 226 Spangle Condos, Zirakpur 226 Spangle Condos, Zirakpur, Mohali, Punjab, India, 140603 Mohali, Punjab 140603

India, Nagaland - Strengthening Nagaland Medical Centre

Developer: Global Himalayan Expedition (GHE) **Project Agreement # PSA** **Date: Q3 2022**

Project Total Cost: US$ 27,777.00 **ISV Seed Funding: US$ 20,000.00** **Self-Funded & Pro-bono: US$ 7,777.00**

PURPOSE

The project involves the solar electrification and installation of critical medical care equipment in the remote Primary Health Centre of Aboi village in the Mon District of Nagaland in India. It will be a first of its kind project where the capacity development will be done at the local level to enable the maintenance of the infrastructure.

The solar electrification of the Aboi Community Health Centre (CHC) will impact the lives of 30 nearby villages and more than 20,000 villagers who are dependent on the CHC for their medical needs.

The lead organization is Global Himalayan Expedition (GHE), a social enterprise that has brought solar energy access to more than 140 villages across 5 regions of India and has also set up several health centres in the remote regions of India.

OBJECTIVE

- Survey of the community health center needs.
- Create a maintenance and management contract for the upkeep of the infrastructure to be put in place.
- Install and commission the solar grid system.
- Encourage the local community through training of the youth, doctors and nursing staff as solar technicians on AC solar grids.
- Create awareness in the nearby villages on the advantages of modern medicine and the upgraded health infrastructure.
- The solar health centre will provide direct economic benefits to local entrepreneurs.
- Benefit the community through ease of access to quality healthcare in a nearby location.

BACKGROUND

The Aboi CHC is located in the Mon District of Nagaland and is one of the oldest Community Health Centre in the region, with a dedicated staff that are serving 1,300 patients on a monthly basis that walk in to the CHC. The CHC caters to 30 villages from the nearby cluster in a radius of 30 kilometres.

The CHC will take over the maintenance and the ownership of the medical equipment once the installation and training is done. From the local team, Dr. Hangsha, who is the Lead Medical Officer at Aboi CHC and his staff will coordinate the project with the support of the DC Mr. Thava.

The resources needed in the form of manpower will be provided by the locals. They will be involved in the local transport and logistics, including the provision of the land for the solar grid system and the structure for the solar panels.

LEVERAGE AND SCALABILITY PLANS

The Solar Electrification of the Health Centres is aimed at improving the overall health care infrastructure in the remote regions of India.

Solution Model: GHE has a list of 10 Health Care Centres that need upgradation in the region. For the same, the strategy is to approach foundations and Corporate Social Responsibility partners who want to invest in improving the health care infrastructure in the remote regions of India. The capital investment for the solar grids and medical equipment will be acquired from the grant funding by GHE.

Sustainability: GHE will be carrying out solar training workshops to train the local youth on setting up both AC and DC solar grids.

Further, a 2-day training program will be provided to the health care workers to operate and maintain the installed electrical and medical equipment. The idea is to build the local capability of the community to handle advanced health care infrastructure providing quality healthcare access.

The infrastructure setup will be handed over to the District Administration and the Medical Department as part of the agreement signed with them, which will ensure the proper use of the investment.

TEAM

- Mr. Jaideep Bansal is the COO of GHE. He has over 10 years of experience in the solar energy and rural development sector and will be coordinating the overall deployment of the project.
- Dr. Hangsha is the Doctor in Charge of the Aboi Health Centre and will be leading the project at the local level.
- Mr. Thava, the Deputy Commissioner of Mon District Nagaland will facilitate with all the necessary approvals and also provide the logistics for the setup of the infrastructure and post deployment.

IMPACT AND TRANSFORMATION

- Total Health Centres Electrified: 1
- Total Lives Impacted: 20,000
- Solar Capacity Installed: 7.5kW
- Total Medical Equipment Installed: 13
- Monthly Patients: 1,300
- Monthly Baby Deliveries enabled: 27

GHE shall provide a detailed Project documentation and quarterly reports to IEEE SV for the project period:

- Documentation of grid specifications
- Impact assessment reports
- Digital images and videos

TIMELINE

- Project Start – Aug' 2022
- Transport of Solar Material- Sep'2022
- Installation of Solar Grids- Oct'2022
- Project End – Dec' 2022
- Training of Local Youth as Solar Engineers- Sep'2022
- Impact Assessment- Dec'2022

AT A GLANCE

Project Locations:	Mon/Nagaland, India
GPS	Latitude: 26.603974E Longitude:94.959695N
IEEE Funding	US$ 20,000.00
Payment Tranches	Tranche 1: US$ 19,935, Tranche 2: US$ 4,984
# local IEEE members involved	2
Number of Beneficiaries (Phase 1)	1 Community Health Centre that will impact the lives of 30 villages and more than 20,000 villagers
Number of Beneficiaries (At scale)	15 Health Centres and 100,000 Lives Impacted
Technologies to be deployed	AC Solar Grid
IEEE Review and Reporting Period	Quarterly and final report
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Jaideep Bansal - jaideep@ghe.co.in
Organization information	Global Himalayan Expedition 226 Spangle Condos, Zirakpur 226 Spangle Condos, Zirakpur, Mohali, Punjab, India, 140603 Mohali, Punjab 140603

Nigeria, Omupo - Electricity for Water and Women Rice Producers' Support

Developer: Shabyis Nigeria Limited **Project Agreement # PSA 054 - 21 Date: Jan 1, 2022**

Project Total Cost: US$ 60,000.00 **ISV Seed Funding: US$ 60,000.00**

PURPOSE

This project deploys renewable energy generation producing electricity for 5,000 residents and the of pumping and distribution of water for 10,000 residents of Omupo Village, Nigeria. It additionally provides productive power for rice processing equipment. This is in alternative the traditional winnowing, debris cleaning and hand washing, substantially helping the community produce high quality commercially acceptable rice. Production of refined and cleaned rice has a ready market commanding a higher price which will dramatically improve the economic base of families throughout the community.

The national grid 33kV feeder supplying electricity to Omupo is in a very poor state. For over 10 years the community finds itself without electricity for extended periods (months at a time) and when power is available it lasts for only about 2 hours at a time.

Shabyis Nigeria Limited (SNL), the developer, is receiving compensation from the Omupo Development Union by providing the facilities to provide sustainable water production to their populace. SNL will engage local labor to construct the facility and install the rice processing equipment and operate the plant after training.

OBJECTIVE

To provide renewable energy based electricity to the residents of Omupo community to enjoy water supply and thus enjoy better quality of life as envisioned by ISV in its mission and vision statements.

BACKGROUND

The project will be located at Omupo or "Omu-ipo," an ancient Igbomina-Yoruba town situated in the southeastern part of Kwara State of Nigeria. It is the nexus of 34 communities of Omupo District, the headquarters of Omupo/Idofian Area Council of the Ifelodun Local Government. The village has a large contingent of migrant women who moved to the area evading the wars and famine taking place about 600 kilometers to the north. They moved down to Omupo and took up the occupation of processing rice. Omupo attracted the migrants due to a 60,000 liters water tank built as part of the United Nations' SDG initiative. Unfortunately, due to the extreme unreliability of power delivery to the village, water pumping into the tank is infrequent. This project will install a 21kW SunBlazer solar system to provide electricity to power the water pump for the 60,000 liters storage tank, relieving its dependency on infrequent and unpredictable grid power. Critically, having sustainable power and consistent water pumping will provide the community a reliable source of clean and hygienic water. This will allow the women rice producers to wash the rice creating a product valued by the community. The power will also drive other rice processing and packaging equipment adding value and ability to sell their products at higher market prices.

LEVERAGE AND SCALABILITY PLANS

The Kabiyesi (King) of Omupo has agreed to provide SNL surveyed land with legal title for the project. ODU (Omupo Descendant Union) executives will help us select trustworthy members of the community to hire/train to installing the equipment and its operation.

Furthermore, ODU is providing us a building for our materials and staff. ODU executives to provide security for our equipment and personnel. ODU will also provide necessary security for SNL's equipment and personnel.

IMPACT AND TRANSFORMATION

This project will immediately impact the life of over 5,000 people as they use the water for domestic and commercial applications.

The reliable potable water along with processing equipment will transform the lives of hundreds of women who produce and process rice directly; and the thousands who will now have access to good quality rice.

The project will be transformative to the Omupo community members driving substantial economic benefit across the region.

TEAM

- Engr. Tunde Y. Salihu, MD/CEO of SNL Lead promoter of projects Mr Shuaibu Ottan, Chairman board of Directors of SNL, a former banker and has been on our projects for 6 years.
- Engr. Kolawole Adewole, Director, SNL, a former development bank executive. He has been involved in our projects for 4 years.
- Alhaji Yinka Yahya, Director SNL, a former banker. He has been involved in our projects for 1 year.
- Alhaji Sola Adefemi, Director SNL, a former banker. He has been involved in our projects for 6 years.
- Engr. Salamat S. Salihu Business Development Director. She has been involved in our projects for 6 years.
- Bilikis Salihu, Finance Manager. She has been involved in our projects for 3 years.
- Engr. Dare Project Manager. Has been involved for 2 years.

TIMELINE

The survey for this project started in April, 2021. Project documents executed in November of 2021. Solar power system and integration with water pumping took place Q1 of 2022. Concurrently, SNL procured and installed the rice milling and packaging machines and materials. Test run and system commissioning is scheduled for early 2nd quarter with commercial rice mill operations commencing late in Q2 2022

AT A GLANCE

Project Locations:	Omupo, Ifelodun LGA, Kwara state, Nigeria
GPS	DMS: 8.2667°N 4.8000°E Coordinates: 8.2667°N 4.8000°E, at an elevation of 1,220 ft (370 m)
IEEE Funding	60,000.00 US$
Payment Tranches	Tranche 1- 65%, Tranche 2: 30%, Tranche 3: 5%
# local IEEE members involved	2
Number of Beneficiaries (Phase 1)	5,000
Number of Beneficiaries (At scale)	10,000
Technologies to be deployed	Solar powered water pumping for rice processing
IEEE Review and Reporting Period	Quarterly and final report
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Tunde Salihu - tunde_salihu@yahoo.com
Organization information	Shaybis Nigeria Ltd 390 Alimi road, Beside Total filling station, Oloje Shaybis Electropreneur Center, TIC, Agric compound, off GSS roundabout Ilorin, Kwara 240272

USA, Arizona - Solar PV Living Lab for Entrepreneurship Incubation Center

Developer: Dine College **Project Agreement # PSA** **Date: Q3, 2022**
Project Total Cost: US$ 44,200.00 **ISV Seed Funding: US$ 44,200.00**

PURPOSE

Proposed is a photovoltaic system designed as a 'Living Laboratory". The Lab will be located at the Center for Entrepreneurship Development (Business Incubator) on the Diné College campus at Tsaile, Arizona, located in the northeast corner of the Navajo Nation. This solar project will provide energy to the campus and site training for students including the public.

It is intended as an example for additional solar installation projects and to encourage small business startups. 1st year goals include providing solar energy training to 150 students and 200 community members resulting in 2 business startups. Diné College and ISV/North American Working Group (NAWG) will work with faculty and department managers to identify the location and implementation logistics.

OBJECTIVE

1. Install a solar PV system micro grid to provide 10kWp (peak power) and about 30-50 kWh/day.
2. 30 students/community members trained in the first quarter on the basic photovoltaic power generation and assessment of system performance.
3. Develop procedures to compliment the curriculum used in training for the "Living Laboratory".
4. Conduct business skill workshops to explore and brainstorm practical uses of the energy providing education to branch out as entrepreneurs.
5. Establish business incubation centers/Micro Economic Zones by enhancing telecom services.

BACKGROUND

Diné College is the first tribally controlled and accredited college in the United States, established in 1968 as Navajo Community College, it is open to the public serving primarily women and youth. Our vision is to create local economic opportunities, led and guided by community members connecting and creating current and future entrepreneurs in alignment with ISV's three pillars of Energy, Education and Enterprise Development.

Extend the training to 200 people including Diné College students and Navajo community members in monitoring the power generation and consumption in real time.

Enhance training to include small business startup and expansion classes as well as introducing the concept of Micro Economic Zones that promote exporting of excess goods produced.

LEVERAGE AND SCALABILITY PLANS

This system would be an integral part of the Living Laboratory for skill development of the entrepreneurs from the Navajo communities. This is to train Navajo youth and women in the area of renewable energy systems and related productive use businesses.

This proposal is for phase one. Additional renewable energy and locally relevant technologies will be added as we build the living laboratory in subsequent phases. Long-term plan is to build a partnership between Diné College PV system and Navajo community.

TEAM

1.James Denney: School of Business/Social Sciences, Diné College. Liaison between the college, the Navajo communities, and the Business Incubator.

2.Dr. Michael Lerma: Dean, School of Business & Social Science at Diné College. Owns launch of a virtual business incubator.

3.Dr. Ashok Das: Founder/CEO, SunMoksha. Expertise in clean technology innovations, microgrids, smart irrigation projects.

4. Adrian Lerma: Alumni Relations, Diné College. Co-founder of Navajo Women's Energy Project (2012) & Native American Business Incubator Network (2014). Expertise in social media/brand/website management, community outreach/grassroots organizing.

IMPACT AND TRANSFORMATION

A new rooftop solar PV system will be installed on the building of the School of Business and Social Studies where the proposed center for entrepreneurial learning will be hosted. The trainees will be the Navajo youth and women from the communities across the Nation. The total anticipated number of trainees is 150 in two years.

TIMELINE

Timeline to be added after aligning with the President of Dine College. Prepared excel sheet already.

AT A GLANCE

Project Locations:	Dine College, North America
GPS	Latitude : 36.2939° N Longitude : 109.2162° W
IEEE Funding	44,200.00 US$
Payment Tranches	Tranche 1- 65%, Tranche 2: 30%, Tranche 3: 5%
# local IEEE members involved	
Number of Beneficiaries (Phase 1)	30
Number of Beneficiaries (At scale)	350
Technologies to be deployed	Solar PV system, adjustable Support Structures (ground mounted and/or rooftop), Storage, Inverter and (BOS) Balance-of-System to convert DC energy produced by solar panels to AC power.
IEEE Review and Reporting Period	Quarterly and final report
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Ashok Das - das@sunmoksha.com
Organization information	Dine College 1 Circle Drive, Route 12 - 1 Circle Drive, Route 12, Tsaile, AZ 86556 Tsaile, AZ 86556

Zambia, Mayukwayukwa - Reliable Solar Energy for an Established Refugee Settlement

Developer: LiChi's Community Solutions Ltd. **Project Agreement # PSA** **Date: Q4 2022**

Project Total Cost: US$ 200,000.00 **ISV Seed Funding: US$ 200,000.00**

PURPOSE

The ultimate goal of this project is to install a micro solar panel grid that will replace use of diesel generator that is not reliable due to shortages of fuel. More information in attached GHTC paper, will be installed in a designated house cluster and will, through proof of implementation success, be scaled to provide power access to the whole camp. As consistently stated, this project will be installed in the Mayukwayukwa refugee camp located in the Western Province of Zambia. This project will aim to target the UN workers, health workers and residents that are already connected to diesel generator and will sell to them the power per kwh in order to enable them to have reliable power supply, for teachers to make work plans, school children to read at night and refrigeration to have power for those with fresh goods to be able to preserve them and sell them fresh. More power will be needed during day for hair salons, internet and other commercial power appliances at the market that will enable people of Mayukwayukwa to create new businesses and methods of stimulating their local economy development. Almost every person interviewed at the time of the on-site visit stated that they either had a business or that, if granted access to a more reliable means of power, they would like to start and maintain a business. The initial system build out and operative period will be completed within two years and reporting and scaling will be continued for five years approximately.

OBJECTIVE

- Technology: The on-site assessments verified that the technology is appropriate and desired by the community. The invitation to Likonge by the ministry indicates both government approval and likely government and/or UN support.
- Entrepreneurship: This is a business that will earn enough to maintain and expand. Although the expansion may be slow but that is usually the case with utilities. There is government and UN interest in the project so there is ample reason to believe that a successful prototype will result in availability of funding. Likonge is on board to find additional funding.
- Education: Likonge and her social enterprise LiChi Community Solutions will educate engineers in the community to maintain the equipment.

BACKGROUND

To truly address the what it is that this concept aims to solve, it is important to consider what it actually means to have a reliable power source. Globally, for an average person with access to electricity and power, this means access to basic human needs such as lighting, water pumps, adequate health care, business potentials, improved methods of education, etc. For the people of Mayukwayukwa, this new beneficial addition to their livelihood would mean all of these things and more. To elaborate, the camp already has in place a bank to support the expected increase in the number of financial transactions. It is important to clarify that this bank is one of the only ones currently established in the Western Province. The people in this camp have reported their plans to open their own businesses as a means to make their income - sewing clothes with machinery, metal work to improve infrastructure, beauty salons, etc. Additionally, they would have the chance to use appliances like freezers to store their food rather so that they may enjoy things like flavorful meat and safe eggs days after purchase (gone will be the days of drying out food as a means of preservation).
Many of these goals, even the most basic of them, can only be accomplished with a sufficient source of power. Giving this community power isn't an exceptional goal, but rather it is the most obvious move in favor of human decency and the right every person holds to improve their lives.

LEVERAGE AND SCALABILITY PLANS

A basic business model proving self-sustainment and sufficient revenue for growth is available. It is known that the families will have no grievance with paying for power received -- collections will serve maintenance purposes. Two possible schematics of the actual micro grid layout are mapped out. This will allow for proof of concept to be made for UN support on expansion.

TEAM

This project will be under the supervision and implementation of Likonge Makai Mulenga who is director of LiChi's Community Solution Ltd (LCS). The LCS Ltd was founded in 2014 in collaboration with Kilowatts for Humanity on Filibaba Project funded by IEEE Smart Village from 2014 to date.

IEEE Smart Village sponsored LCS with solar lighting kits for Shikwakala and Filibaba in 2015. The Mayukwayukwa project is been supported by director of CU Boulder Electrical Engineering Professor Alan Mickelson who also serves as the Director of the Operations Committee for IEEE Smart Village from 2018 with the survey and location evaluation using two students. LCS has 8 organization staff, 5 volunteers and 6 board members.

IMPACT AND TRANSFORMATION

Within five years, this project will impact every camp resident, refugee, and worker. This totals roughly 15,000 individuals. More than this, it is undeniable that when considering the potential economic and business opportunities established as a result of this installation, the true figure of individuals likely impacted is much higher. It is plausible to assume, that due to the elevated business status of the camp area, the entirety of the Western Province (pop. 902,000) will be impacted.

TIMELINE

Timeline to be added after confirmation

AT A GLANCE

Project Locations:	Zambia - Mayukwayukwa
GPS	-14.539071239551264, 24.207747200866237
IEEE Funding	200,000.00 US$
Payment Tranches	Tranche 1- 65%, Tranche 2: 30%, Tranche 3: 5%
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	15,000
Number of Beneficiaries (At scale)	902,000
Technologies to be deployed	Solar PV System
IEEE Review and Reporting Period	Quarterly and final report
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Makai Likonge - likobus@yahoo.com
Organization information	LiChi's Community Solution Limited 12159 Chingola Central, Off Kabundi Road Hombwe Avenue, House No. 234, Mufulira Zambia. Chingola, 10101

India, Odisha - Solar Smart Irrigation for Kudagaon Island Village

Developer: Sun Moksha **Project Agreement # PSA 063 - 22 Date: Q3 2022**
Project Total Cost: US$ 31,276.00 **ISV Seed Funding: US$ 24,984.00** **In-kind support: US$ 6,292.00**

PURPOSE

Solar Smart Irrigation project will provide reliable, sustainable and scalable irrigation system for farmers. SunMoksha's Smart AQUAnet™ will enable farmers adapt to data-driven, scientific, precision irrigation. It optimizes water usage based on crop, soil type & soil conditions. It will create awareness & build capacity for smart irrigation & precision agriculture.

The project will be implemented at Kudagaon Island on Mahanadi River in Odisha, India. It will directly benefit 192 people & increase their income by >2X within a year.It will create auxiliary livelihood and entrepreneurship opportunities: support-labor, market linkage, food processing, etc. SunMoksha has been working in Kudagaon since 2018 when it was implementing its microgrid solution "Smart Nanogrid™" for reliable power.

OBJECTIVE

The objective is to increase the income of the farmers through a scientific, data-driven approach for agriculture. The project aims to:

1. Move 48 farmers from 1 crop to 3 crops & increase their income by >2X.
2. Introduce them to scientific farming & precision irrigation to optimise water.
3. Introduce farmers to beneficial cropping patterns
4. Develop a sustainable business model.
5. Generate data on crop vs. water consumption for various crops grown in that region.
6. Educate the farmers towards climate adaptation.

Expected impact of the project is immediate & can be seen within a year such as:

1. Increase in farmer's income by at least 2X.
2. Adoption of precision irrigation methods to optimise water requirements.
3. A self-sustainable business model managed by the community.

BACKGROUND

- Kudagaon is an Island on river Mahanadi in Odisha, India (Population of 350 people, 85 households). Villagers' main occupation is agriculture, practised only 4 months during the monsoons. They work on daily wages as masons or labourers in nearby villages and towns, for the rest of the year. Major crops grown are rice, sweet potato, eggplant, groundnuts, occasionally tomato & wheat.
- SunMoksha has been working in this community since 2018. It brought electricity for the first time through a 20 kWp solar Smart Nanogrid™ in 2019 which is operated and maintained by the elected Village Energy Committee (VEC).
- SunMoksha is now working to bring irrigation water to the community, which wants to do cultivation throughout the year and are eager to learn scientific and precision irrigation.
- It is proposed to elect a "Village Water Committee" (VWC) to own and operate Smart AQUAnet™ with AMC from SunMoksha.
- Land will be provided by the community for extending microgrid and installing AQUAnet™.

LEVERAGE AND SCALABILITY PLANS

Total quantity of water that will be pumped every day is approximately 150,000L, 5hr of operations. Scheduling features of Smart AQUAnet™ will be used to schedule the distribution on a daily/weekly basis. Billing will be done based on actual consumption by each farmer. We plan to charge INR 100 for 10,000L of water. A revenue of INR 1500 will be collected every day for approximately 8 months. The total revenue expected in the 1st year is INR 83,200/-. This net revenue will be deposited in a new account which will be created by the Village Energy Committee (VEC) at Athamalik, which is a registered entity. This new account will be used for all transactions to be done by the Village Water Committee (VWC).

TEAM

- Dr. Ashok Das, Founder CEO; SunMoksha. 25+ years in Renewable Energy, smart microgrid, IoT, smart irrigation, rural electrification & smart villages.
- Ayushi Sharma; research & community engagement. Development of knowledge repository of Smart Nanogrid™ & Smart AQUAnet™. Post implementation, she conducts impact assessment.
- Ameet Deshpande; 10+ years of experience in microgrid & irrigation solutions, and has implemented such projects in the villages.
- Elected Village water Committee (VWC) and VEC.

IMPACT AND TRANSFORMATION

The system will deliver 150,000 LPD from 11.5kWp of solar PV to ensure water availability throughout the year. It will directly benefit 48 farmer families (192 people) & increase their income by >2X within a year. It will create awareness & build capacity for smart irrigation & precision agriculture. It will create opportunities for auxiliary livelihood and entrepreneurship in the village, such as, support-labor, food processing, market linkage, etc. The current irrigation project is the next step in our intervention. The goal is to develop a sustainable smart village. In the next phase we will expand the smart irrigation project to add remaining farmers & build a Smart Micro-Economic Zone (Smart MEZ™) to add value to the produce, including fisheries and other energy-enabled livelihood.

TIMELINE

Project start and end date: Start Date: February 2022, End Date: May 2022
Key milestones start and end dates: Creation of sump – Feb 2022, setting of plant: Mar 2022, setting of pipeline: April 2022, training and capacity building: May 2022

AT A GLANCE

Project Locations:	Kudagaon Village, Athmallik Tehsil, Angul District, Odisha, India 759125
GPS	Latitude: 20.737893˚N Longitude:84.4715593˚E
IEEE Funding	24,984.00 US$
Payment Tranches	Tranche1: 12,492 US$, Tranche2: 9,994 US$, Tranche3: 2,498 US
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	48 marginalized farmer families (192 people) will be direct beneficiaries
Number of Beneficiaries (At scale)	85 families (350 people) will be direct beneficiaries. Neighbouring 100 families indirect beneficiaries
Technologies to be deployed	IoT, Cloud, AI, Solar pumping systems, SPV to extend microgrid capacity
IEEE Review and Reporting Period	Monthly
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	Dr. Ashok Das - das@sunmoksha.com
Organization information	SunMoksha Power Private Limited B9-602 South City, Arekere Micro Layout B9-602 South City, Arekere Micro Layout, Bangalore, Karnataka, 560076, India Bangalore, Karnataka 560076

Argentina, Chaco - Electricity to Combat Deforestation Effects

Developer: Monte Adentro **Project Agreement # PSA 064 - 22 Date: Q4 2022**
Project Total Cost: US$ 200,000.00 **ISV Seed Funding: US$ 200,000.00**

PURPOSE

Residents from El Impenetrable in the Chaco province suffer from extreme poverty, hunger, and disease. They are increasingly leaving their lands in search for opportunity elsewhere, leaving the region vulnerable to exploitation and deforestation. The objective of this project is to use solar energy to improve livelihoods and uplift communities where opportunities are scarce.

Solar PV microgrids will power basic household services and solar water pumps, while electricity in community centers will expand Monte Adentro's professional and school support programs, and promote productive uses of the services. In this way, the project improves household living conditions, alleviates hunger and water insecurity, and fosters micro-entrepreneurship and education. The project will serve a total of 970 people or 210 households living across seven rural communities.

Monte Adentro is the organization overseeing the entire project, with two university engineering groups supporting the design and implementation phases. During implementation, local students from a nearby technical school will also receive training to support the project and further their own education. When the project is established, a local solar cooperative will partner with Monte Adentro for regular O&M. The implementation is anticipated to last 12 months from the start of construction, with continuous Monitoring and Evaluation for the five following years.

OBJECTIVE

This project aligns closely with the three pillars of education, entrepreneurship, and technology by (1) enabling improved access to educational support and professional training, as well as providing local technical training during implementation, (2) providing fundamental household and community services powered by solar microgrids, and (3) strengthening micro-entrepreneurial opportunities for community members.
This project was born out of the communities' expressed desire for it and has been designed with their input. There is also the involvement of other local stakeholders, including the EET20 Technical School in a nearby town and solar cooperatives working in this region.

BACKGROUND

The Chaco region in Argentina is one of the most underserved of the country, often in the spotlight for its high levels of poverty, water and food insecurity, and low levels of education, employment and health services. Additionally, it has one of the world's highest deforestation rates due to uncontrolled resource exploitation and expanding agriculture, leading to environmental degradation, now heightened by climate change.

The rural and indigenous communities of the region are vulnerable due to the geographic and socioeconomic exclusion they face, as is common worldwide. They face two fundamental challenges: access to basic needs for survival and sufficient health; and resources for economic productivity. They are facing increasing food and water shortages that cause malnutrition, water-related diseases and low levels of economic productivity. This project takes Monte Adentro's vision and mission of allowing the communities to realize their full potential by enhancing their existing capacities and incorporating physical solutions to material needs that the communities cannot address by themselves.

This project is designing solutions for long-lasting adaptations, leveraging energy access to improve day to day living, small-scale food production and storage, water and sanitation access, and entrepreneurial opportunity.

LEVERAGE AND SCALABILITY PLANS

The benefiting families are able and willing to pay fees for energy and water services, which will be managed by meters at adequate costs. This project is expected to generate US$ 2, 000 of income every month to make it self-sustaining. Project growth will be driven by other grants to expand infrastructures, such as USAID's DIV, SPA, or CDP programs.

TEAM

- Kellen Kennedy - Undergraduate Research Assistant at University of Colorado Boulder - 1 year.
- Paula Perez - Undergraduate Research Assistant at University of Colorado Boulder - 1 year.
- Alan Mickelson - Professor at University of Colorado Boulder.
- Guillermo Catuogno - Professor at Universidad Nacional de San Luis.
- Martin Gabriel Lucero Menez - Undergraduate Research Assistant at Universidad Nacional de San Luis.
- Elias Andrada - Undergraduate Research Assistant at Universidad Nacional de San Luis.
- Juan Chalbaud - Monte Adentro Liaison.

IMPACT AND TRANSFORMATION

By expanding entrepreneurial and professional opportunities, and improving living conditions and health (reducing time and monetary costs), this project expects households to see an increase in income.

The project is self-sustaining and designed so it can be expanded within the communities and beyond. There is potential for scale due to Monte Adentro's expanding influence, and a market size of over 1.4 million rural, underserved people in the northern provinces of Argentina, around Chaco.

TIMELINE

Project start and end date subject to funding availability.

AT A GLANCE

Project Locations:	Chaco, Argentina
GPS	Quemado Chico -26.2097895378, -60.567895658
IEEE Funding	US$ 200,000.00
Payment Tranches	Tranche1: US$ 12,492, Tranche2: US$ 9,994, Tranche3: US$ 2,498
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	48 marginalized farmer families (192 people) will be direct beneficiaries
Number of Beneficiaries (At scale)	85 families (350 people) will be direct beneficiaries. Neighbouring 100 families indirect beneficiaries
Technologies to be deployed	IoT, Cloud, AI, Solar pumping systems, SPV to extend microgrid capacity
IEEE Review and Reporting Period	Monthly
Sustainability/ Scalability	Project will be sustainable after 1 year
Project Contact	
Organization information	Monte Adentro NA Pampa del Indio, Chaco

Tanzania - Iringa - Honey Farming Hub

Project Agreement # Date

Project Total Cost: US$ 26,000.00 **ISV Seed Funding: US$ 18,000.00** **In-kind support: US$ 8,000.00**

PURPOSE

The goal of the project is to establish a honey farm using modern methods at a Tanzanian NGO Rural Development Organization (RDO). It is part of a larger program for providing electricity to the communities, with honey generating income to help sustain the program. The completed project will provide model that other organizations can adopt to implement their own honey farming projects.

The project will be implemented in the districts of Mufindi and Kilolo. Since 2001 RDO has been actively engaging the communities in these areas through water supply, vocational training and a home-based orphans support program. In phase 1 of the project a group of local beekeepers (RDO and farmer unions) will be provided with education on beekeeping using a Training of Trainers approach.

OBJECTIVE

Milestone 1
- Objective: Expert beekeeper to provide a 4-week training to local beekeepers.
- Impact: Local beekeepers to learn modern beekeeping methods.

Milestone 2
- Objective: Setting up honey production at 2 sites (with different environmental conditions) with guidance by the expert beekeeper.
- Impact: Honey production sites operational, serving as pilot for its operation.

Milestone 3
- Objective: Complete one honey production (harvesting) and processing cycle, remotely guided by the expert beekeeper.
- Impact: Local beekeepers to be fully familiar with all honey farming activities and capable to train others; production sites to produce honey.

Milestone 4
- Objective: Project evaluation and development of training curriculum.
- Impact: Project impact can be assessed; results become replicable.

BACKGROUND

The communities in rural Tanzania rely mainly on subsistence agriculture for securing their livelihoods. Honey is a product which is in high demand both regionally and on the international market, and the environmental conditions in these rural areas favor its production. Income from honey sales can catalyze rural economic development.
The success of honey farming in the rural areas is held back by the following barriers:
- lack of access to knowledge about honey farming
- lack of access to high-quality honey farming equipment
- lack of access to markets
- lack of access to finance to invest into a honey farming business

In phase 1 of the project, we will address the knowledge and equipment barriers by involving an expert beekeeper who will provide education and by establishing ways for obtaining the required equipment. These activities will involve beekeepers employed by RDO as well as beekeepers reached through existing farmer unions facilitated through RDO programs.

LEVERAGE AND SCALABILITY PLANS

The development of honey farming know-how at RDO will act complementary to RDO's sound foundations towards achieving commercial success and to become capable of sharing the knowledge with others.
Project execution will be managed by the project manager and controlled by the RDO steering committee. Project outcomes assessment using indicators and risk management will be defined as part of the detailed project plan.
After phase 1 of the project, the initiative will be scaled up based on (1) the knowledge gathered, and (2) subsequent financial support.
RDO already has access to markets to sell the produced honey to sustain the project. Local farmers will be supported by training and access to equipment, and they can sell their raw honey to RDO who will in turn process it and sell it.
We will develop guidelines for setting up honey production sites and a training curriculum for the purpose of replicating the results for further communities engaged by RDO as well as other ISV organizations.

By expanding entrepreneurial and professional opportunities, and improving living conditions and health (reducing time and monetary costs), this project expects households to see an increase in income.

TEAM

- Daniel Oduntan, originally from Nigeria, is a beekeeper with 29 years of experience and author of beekeeping textbooks. He will provide training and technical support.
- Lawrencia Paul has been managing the RDO beekeeping program since 2020. She is the technical project manager.
- Fidelis Filipatali is the director of RDO. He is responsible for funds management and successful project execution.
- Michael Luger has been with RDO since 2017. He is responsible for project reporting.

IMPACT AND TRANSFORMATION

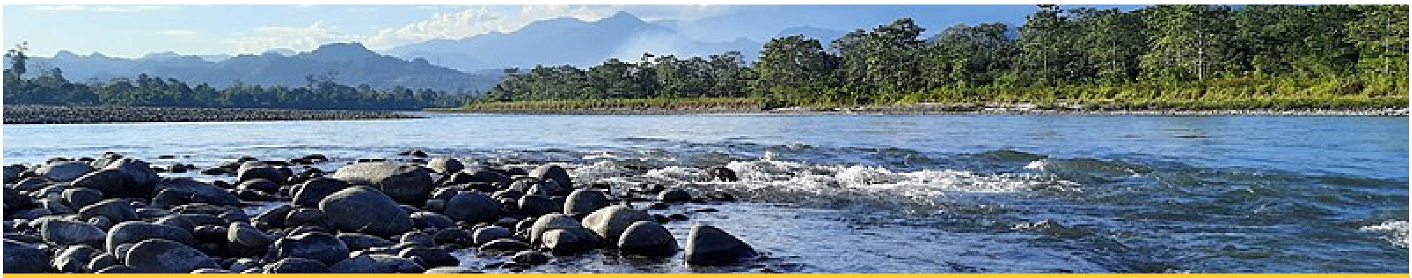
- 10 beekeepers will be trained in modern beekeeping methods, so they become capable of training others.
- 2 pilot sites with 30 hives each, and a processing center will be set up.
- By the end of the project, we expect to harvest 500 kg of honey.
- A set of guidelines and a training curriculum for honey farming in Africa will be developed which can be used replicate the project results.

TIMELINE

- Project start: 21 Mar 2022 (tentative) - Project end: 20 Dec 2022 (tentative)
- Milestone 1 - Training: 21 Mar - 20 Apr 2022
- Milestone 2 - Site setup: 16 Apr -15 May 2022
- Milestone 3 - Honey farming activities: 16 May - 20 May 2022
- Milestone 4 - Evaluation and curriculum development: 21 Mar - 20 Dec 2022

AT A GLANCE

Project Locations:	Mufindi and Kilolo districts, Iringa region, Tanzania
GPS	Latitude: 8°29'59.43"S Longitude: 35°30'53.88"E
IEEE Funding	US$ 18,000.00
Payment Tranches	Tranche 1: US$ 18,000
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	10 trained beekeepers capable of doing modern honey farming and providing training to others.
Number of Beneficiaries (At scale)	100+ beekeepers in East Africa reached by trainings. 37500 people benefiting indirectly through RDO/ISV program "Rural Electrification & Education"
Technologies to be deployed	Modern beekeeping methods and equipment: Langstroth hives, queen bee rearing, advanced colony management.
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	Project is for for technical advancement, surveys and other opportunities
Project Contact	Michael Luger - michael.luger@gmail.com
Organization information	Rural Development Organization (RDO) P.O. Box 65 Mafinga,



India, Assam - Sustainable Rural Community Development

Developer: Tezpur University **Project Agreement # TBA**
Project Total Cost: US$ 60,688 **ISV Seed Funding: US$ 52,688.00** **In-kind support: US$ 8,000.00**

PURPOSE

Project goal is to educate and train rural communities through the creation of a learning platform that disseminates useful applications of renewable energy in sustainable farming, agricultural processing, and modern aquaculture technologies. Project location: Jhawani-3 Village, Bihaguri development block, Sonitpur District of Assam, India. The Jhawani-3 Village will provide participating members while nearby rural communities will be benefit learning about the project for further replication. The anticipated duration of the first phase of the project is 18 months. This encompasses the installation and commissioning of the hardware and facilities.

The project includes: integrated solar irrigation, agricultural-processing with cold storage facilities, and introduction of modern aquaculture technologies. Training encompasses all operational aspects of the program for a further 18 months. The goal of the three year program - to demonstrate a sustainable business model that brings sustainable and transformative economic development and education to the community. The project brings positive impact enhancing the living standard of the villagers with the introduction of several cutting-edge integrated and symbiotic processes with field crop and fish based farming. Scale is achieved with the expansion of entrepreneurial benefits to a large section of village population

OBJECTIVE

The project is directly addressing the three pillars of the ISV mission statement: Energy, Entrepreneurship and Education. Indirectly this will have a positive impact on the general education and welfare of the community. The entire process will be planned through a community participatory mode, augmenting their inherent skill of farming (crop production and processing) and fish production, enabling them to conserve resources (water and energy) and bolster productivity and income.

The support of local academic institutes (Tezpur University and ICAR-CIFRI Guwahati Regional Centre), ISV, Government Departments (part of the monitoring and advisory committee) and similar organizations are expected.

BACKGROUND

Developmental issues, including health, education, roads, communication, and modernization of farming exist despite the fertile land and hard-working rural population. Crop yields remain low primarily due to dependence on rainfall and the unaffordable operating costs of bore wells. Further, there is a lack of local processing facilities for crop transformation, contributing to value losses on farm produce.

This project demonstrates practical solutions to fish aquaculture with the introduction of new techniques and practices. It builds local capacity, inclusive of rural practices with a highly participatory modal. The project planning integrates neighboring villages and motivates the entire region with an expanded range of products produced in an economical and sustainable manner, transforming agrarian development in the entire region.

LEVERAGE AND SCALABILITY PLANS

Capital Cost of the project is US$38,649 whereas operational cost for the 1st year is US$17,716. The anticipated revenue from services of agro-processing units, irrigation and fishery is US$30,759 in the 1st year.

With 8% rate of discount, US$26,408 is estimated as NPV at the end of 5 years based on the revenue and the operational cost. Five years accumulated NPV of profit is estimated as US$33, 069 which will be available for scaling up of the project including further expansion.

TEAM

- Prof V K Jain, Vice Chancellor
- Professor Dhruba Kumar Bhattacharyya, Pro-Vice Chancellor
- Dr Biren Das, Registrar

Prof S K Sinha, Dean, School of Engineering (There are three other Schools headed by three Deans)

Department of Energy has the following members:

- Prof Debendra Chandra Baruah, Professor & Head
- Prof Dhanapati Deka, Professor
- Prof Rupam Kataki, Professor
- Dr Sadhan Mahapatra, Associate Professor
- Dr Pradyumna Kumar Choudhury, Assistant Professor
- Dr Nabin Sarmah, Assistant Professor
- Dr Biraj Kumar Kakati, Assistant Professor
- Dr Vikas Verma, Assistant Professor

There are 27 academic Departments with 261 Faculty Members.

IMPACT AND TRANSFORMATION

Impact anticipated:

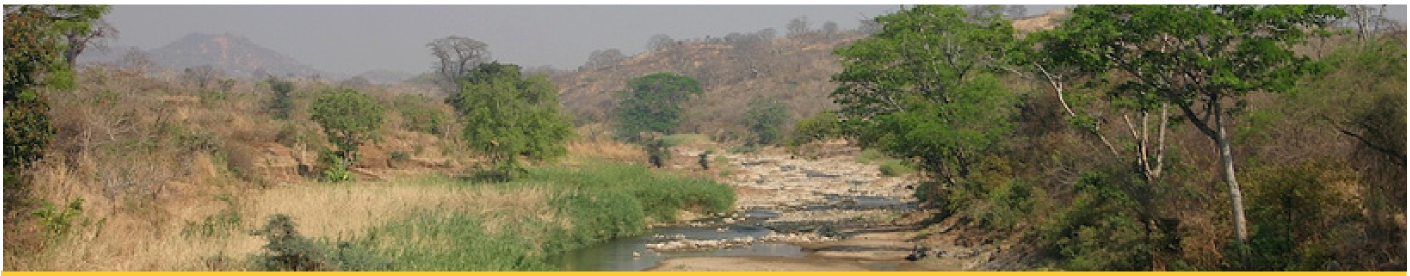
1. Increase in farm productivity due to assured irrigation
2. Enhanced farm income due to assured value addition at farm level (agro-processing unit)
3. Higher family income through management of water bodies through technologically sound sustainable fish farming.
4. Generation of entrepreneurship opportunities for an existing cooperative society run by the villagers.
5. In the long term the project is projected to generate a revenue of US$34,058 with margin of US$27,561.

TIMELINE

Project initiation - Q3 of 2022 with execution over the following year.

AT A GLANCE

Project Locations:	Village cluster Jhawani3
GPS	Latitude: 26°38'18.6"N, 26°38'16.0"N Longitude: 92°41'20.1"E, 92°41'34.2"E
IEEE Funding	US$ 52,688.00
Payment Tranches	N/A
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	261 faculty members
Number of Beneficiaries (At scale)	Faculty students
Technologies to be deployed	Education learning pltaform
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	In the long term the project is projected to generate a revenue of US$34,058 with margin of US$27,561.
Project Contact	Prof Debendra Chandra Baruah - baruahdeben@gmail.com
Organization information	Tezpur University Tezpur University Napaam, Tezpur Sonitpur, Assam (India) Pin - 784028



Malawi, Lilongwe - Fish for Incomes and Sustainable Homes (FISH)

Project Agreement #   Date
Project Total Cost: US$ 29,840.00   ISV Seed Funding: US$ 17,928.00   Self-funded and Pro-bono: US$ 11,911.00

PURPOSE

FISH is a pilot project intended to develop fish farming for commercial markets and for home consumption in the community of Lilongwe, Malawi. Utilizing the proposed solar powered Aquaculture Centre, FISH will supply fingerlings and train local farmers in aquaculture business practices in addition to assisting farmers with the processing, distribution, and marketing of the harvested fish.

Through the commitments of approved farmers, FISH will expand from an initial group of 5 farmers into a cooperative of 100 commercial fish farming operations after 5 years. Edgar Kapiza Bayani (ISV member since 2016) founder of WETECH Ltd., is the project lead. Utilizing the Aquaculture Centre will ensure the ability to recruit, train and develop entrepreneurship in the region. Supporting fish farmers with education and guidance will enhance production, quality, and profitability.

OBJECTIVE

1. Recruit an Intern to help with full time management of Aquaculture Centre.
2. Identify and lease 1 hectare of land for 5 years to set up Aquaculture Centre.
3. Purchase and covert 20ft shipping container to be used as the Aquaculture Centre Office and powered by solar energy.
4. Purchase and install a solar water pump to support for fishponds.
5. Construct 2 fishponds stocked with 4,800 Tilapia fingerings.
6. Identify initial 5 farmers, support the installation of 1 fishpond per farmer.
7. Stock each farmer's pond with 2,400 Tilapia fingerings.
8. Train farmers in fish farming practices (management, feed, harvesting, and marketing).
9. Organize aggregated marketing and distribution of fish. Successful fish farm projects can be replicated and supported by the Aquaculture Centre.

BACKGROUND

Project targets farmers with suitable land for fish ponds in the peripherals of Lilongwe City, Malawi's capital.
These farmers are the primary suppliers of food items to Lilongwe city residents.
Edgar, an IEEE/ISV member leads the project under WETECH Ltd., was trained in Fish Farming .
Edgar will be assisted by Vincent Kamwanya and Henry Mapwesera. Both are fisheries experts with a combined 20+ years of experience.
Three farmers have already been identified on criteria having ownership of land, willingness to contribute and sell fish in a joint effort. Farmers will contribute the labor for excavation of ponds & MoU to be signed before project support. Project will remotely be supported by other ISV members like Earnest Nge, Rajan Kapur and Abdulateef Aliyu.

LEVERAGE AND SCALABILITY PLANS

With the support of local seasoned Fisheries experts Vincent Kamwanya and Henry Mapwesera, Edgar will lead the Aquaponics Centre to be self-sustained through the sale of fish fingerlings, and training of new farmers.

After the initial 5 farmers, FISH will continue to support at least an additional 10 farmers each year with the aim to reach 100 farmers after 5 years.

Edgar will do designs and installations of solar installations (water pump and power for 40ft converted container) as an additional measure to continue growth efforts.

TEAM

1. Edgar Kapiza Bayani, WETECH CEO is lead. Edgar an energy entrepreneur with a background in agriculture and natural resources management was trained in Aquaculture when he worked for USAID/Chia Lagoon WaterShed Management and supported the Fisheries Coordinator to 50 farmers including establishing fish farms.
2. Vincent Kamwanya (working for Kumudzi Kuwale) and Henry Mapwesera (working for GIZ) are local fisheries experts to support the project.

IMPACT AND TRANSFORMATION

Long Term Outcome: Increased incomes, access to fish farming products for sustainable pond fish farming populations in and around the Lilongwe Rural/City. Intermediate Outcomes: Increased incomes for 100 farming families around Lilongwe in 5 years.

Improved nutrition for the families in Lilongwe with access to fish and fish products in 5 years. Outputs: Solar powered Aquaculture Centre set up and in operation. Fish Farming "Pass On" Program piloted with 5 Farmers will target an additional 100 farmers within 5 years. FISH Project Business Model developed in alignment with the mission and vision of ISV. Energy: Solar power for water pumping and powering Centre. Entrepreneurship: Increase farmer incomes through fish farming. Education: Aquaculture Centre will educate farmers in fish farming practices and business management.

TIMELINE

Project Start: Q3 of 2022 Key Milestones: In Q3 of 2022: (1) Secure land lease, (2) Recruit Fisheries Intern, (3) Purchase/convert 20ft container, (4) Construct and stock 2 fishponds with 4,800 fingerlings, and (5) Install solar powered pump systems accordingly. In Q4 of 2022, Recruit, train, and support 5 farmers in pilot. In 2023, perfect business model.

AT A GLANCE

Project Locations:	Lilongwe, Malawi
GPS	-13.885825269159536, 33.77192008145748
IEEE Funding	US$ 17,928.00
Payment Tranches	Tranche 1: US$10,757 (60%): Tranche 2: US$6,774 (35%): Tranche 3: (5% after final report): US$896
# local IEEE members involved	1
Number of Beneficiaries (Phase 1)	30 people from 5 Fish farming families
Number of Beneficiaries (At scale)	100 farmers as direct beneficiaries with 600 indirect beneficiaries
Technologies to be deployed	Solar for water pumping and powering office (lighting and computers)
IEEE Review and Reporting Period	Quarterly
Sustainability/ Scalability	2 years
Project Contact	Edgar Kapiza Bayani - wetechmalawiltd@gmail.com
Organization information	Waste and Energy Technologies House No. 178, Ntcheu Street Area 15 Post Box 30345 Lilongwe,

Chile - AgriPV system to Cope with more than 10 years of Drought

Project Agreement #   Date

Project Total Cost: US$ 50,000.00   **ISV Seed Funding: US$ 25,000.00**   **Self-funded and Pro-bono: US$ 35,000.00**

PURPOSE

The project introduces solar power energy increasing the productive capacity of agricultural land where the PV panels are elevated above the growing space.

The goal is to increase land use efficiency while promoting synergy between both systems. Project is located in remote areas of Chile experiencing extended drought and extreme weather events for more than 10 years.

Structurally elevated (3-4 meters) PV modules allow the cultivation of tall-growth produce and fruit trees underneath, with the benefit of shading.

Controlling the solar radiation on fruit, the decrease of crop transpiration and humidity maintenance/water savings, the control of extreme weather events, along with solar powered irrigation system will materially increase crop production.

OBJECTIVE

To radically improve water management, crops performance and quality of life through successful integration of agriculture and PV solution in a Rio Hurtado community.

Specific objectives:

1.To have a long-term AgriPV System and a demonstrative project for the future replication.

2.To take advantage of the impact of the shade and energy use of a solar photovoltaic plant (PV panels) on fruit trees: water consumption, sun shocks, physiological disorder, others.

3.Improve irrigation with photovoltaic energy: water physiological demand, water-energy-food nexus, and to exploit the advantages in the face of extreme climatic events: energy security, water security, food security, resilience to climatic events.

BACKGROUND

The pilot project is located in Río Hurtado, Coquimbo region, with a community of 357, all small-scale farmers. Rio Hurtado has an Inca origin (year 1472) before the arrival of the Spanish colonizers. Previously, since the third century, Molles and Higuitas cultures inhabited the territory that to this day maintains a rural tradition. The main activity of its inhabitants is small-scale agriculture. Since 2010, Río Hurtado has experienced a severe decrease in rainfall as see in the figure (ppt). The University of Chile, Faculty of Agricultural Sciences established a cooperative relationship with the community 2010. The faculty carried out several projects related to crop improvement and the introduction of mechanical irrigation systems. The proposal will select, together with the local municipality, - a small group of small-scale farmers to field test the first pilot. The basic selection criteria should be the impact on their quality of life and the maximization of the impact of the solution. In addition, direct cooperation from the community is expected for the implementation/operation/maintenance of the system.

LEVERAGE AND SCALABILITY PLANS

The PV system can benefit products through the control of ambient temperature, unwanted interference of vegetation, albedo in the case of bifacial technologies, eventual damage by vandalism, use of cleaning routines that allow its optimal performance.

The project will play a dual role: on the one hand, it seeks to have a direct impact on the community through more and better agricultural products, along with access to solar energy. On the other hand, it is a demonstration project that, together with the local municipality, will verify the costs involved, the benefits and learning necessary for maintenance and operation.

Through the municipality, support will be sought for new projects based on the experience gained. The results will be disseminated in order to add new partners and open financing opportunities.

TEAM

- Prof. Dr.-Ing. Rodrigo Palma-Behnke, Director, Energy Center, Universidad de Chile. Senior Member IEEE, Past President Chile Section, PES member.
- Marcia Montedonico, MSc., Agricultural Engineer, Socio environmental specialist, Energy Center, Universidad de Chile.
- IEEE student volunteers.
- Prof. Dr. Mg. Agricultural Engineer Marco Garrido, Director, CEZA Universidad de Chile. - Municipality of Río Hurtado, Coquimbo: 287 farmers (95% small-scale producers) and 70 agro-industrial producer.

IMPACT AND TRANSFORMATION

The AgriPV system will be 3 kW, with a surface area of about 30 m2 (estimated crop area), an average daily production of 15 kWh, with a 30 kWh battery system, a 3 kW single-phase inverter, a 1 kW pumping system and a lighting system of 200 W in total, along with the rest of the electrical consumption in the house.

BENEFITS
- Economic benefits of the project: energy cost, fruit and crop yield, carbon footprint, water footprint.
- Sense of security and protection from the climate situation and drought.
- General improvement of the quality of life in the community.

TIMELINE

Project start on first quarter (Q1) of 2022 and end on second quarter (Q2) of 2023 (18 months)
Key milestones:
- Characterization of the selected location together with the municipality and the community.
- Design and implementation of technological solution designed jointly.
- Commissioning, measurements, analysis and adjustments.
- Comprehensive economic evaluation.
- Adjustment of the design and packaging of the solution (replication, scalability).
- Collection, training and dissemination.

AT A GLANCE

Project Locations:	Río Hurtado, Coquimbo, Chile
GPS	Latitude: 30°16’0” S Longitude:70°40’0” W
IEEE Funding	US$ 25,000.00
Payment Tranches	Tranche1: US$15,000, Tranche 2: US$10,000
# local IEEE members involved	8
Number of Beneficiaries (Phase 1)	Community of Rio Hurtado, near 20 farmers
Number of Beneficiaries (At scale)	Community of Rio Hurtado, near 350 farmers
Technologies to be deployed	PV solution, battery storage, inverter, pump, measurement system, crop monitoring system.
IEEE Review and Reporting Period	Every 6 months
Sustainability/ Scalability	2 years
Project Contact	Rodrigo Palma - rrodpalma@gmail.com
Organization information	Energy Center University of Chile Plaza Ercilla 874 Santiago, RM, Chile 8370450 Santiago, RM 8370450

Darway Coast Nig Ltd – Nigeria – Scalable Minigrids

Project Agreement #   Date: TBAD

Project Total Cost: US$ 624,257.00   ISV Funding Sought: US$ 200,000.00   Self-funded: US$ 424,257.00

PURPOSE

Darway Coast Nigeria Ltd. ("Darway") is a renewable energy provider, specializing in the development of scalable mini-grids and micro-grids for off-grid and underserved areas.
The company's vision and strategy are centred around the provision of clean, sustainable and cost-effective energy to wholly off-grid or underserved communities, SMEs, and commercial and industrial customers. Founded in 2015, Darway currently has three operational off-grid energy facilities. The company expects to develop and implement over 3MW of solar hybrid mini grid projects over the next 2 years.

BACKGROUND AND OBJECTIVE

In response to the clear supply and access constraints in the Nigerian Electricity Supply Industry, Darway identified two primary categories of off-grid initiatives through which it intends to provide electric power to its potential customers. The categories which follow below may also be classified in terms of location, customer segment and capacity.
i.Off-grid mini grids
ii.Interconnected urban/peri-urban mini grids
Darway plans to take advantage of the subsidies available in the Nigerian off-grid electricity sector to support the development of the rural mini grid projects (which as at date still require subsidies to bridge viability gaps). Under the Nigeria Electrification Project (NEP), the World Bank is providing US$80 million in results-based grants which Darway will be leveraging to develop its mini grids.
Based on Darway's identified categories of off-grid initiatives, the company expects to develop and construct over 100 off-grid mini grids in the next 4-5 years.
Darway also plans to provide affordable internet access, electric transportation and agricultural processing services in these beneficiary communities.
Darway by this proposal aspires to develop two minigrids, each with100kW capacity and 256kWh of Lithium Storage in Ihuaba, Rivers State and Garki, Abia State respectively. This project is part of a larger portfolio of 35 free-standing minigrids with a total projected PV output of 5.2MWp with 10MWh Lithium battery energy storage. Upon completion, this phase of project will create over 3,000 new connections and improve the livelihood of over 20,000 people.

LEVERAGE AND SCALABILITY PLANS

We identified over 35 off-grid communities with a potential of over 30,000 connections addressing a population of over 150,000. These communities have no near-term electrification plans by the local utility for the next 5-10 years.

The community comprises of residential, commercial, light manufacturing, public, and anchor customers that are currently powered by other alternative forms of energy such as kerosene lamps, candles, petrol or diesel generators. Based on our survey, the communities are very interested in reliable, renewable, and affordable source of electricity due to the exorbitant cost of operating and maintaining their individual gensets.

TEAM

Henry Ureh (Team Lead): He earned his Bachelors and Master's degree in Electrical Engineering from Cal Poly, San Luis Obispo, California and has over 9 years of experience in utility industry and consulting. Henry will bring his expertise in Renewable energy design and project management to successfully complete these projects. He has designed and engineered PV systems of over 20 MW capacity.

Stanley Onyelusi (Project Lead): With extensive experience in solar installation and the project management large scale renewable deployments, Stanley has commissioned over 3MW of PV projects, both in residential and commercial settings. For the pilot projects he will oversee the feasibility studies, load audit, preliminary design, cost estimation, and engineering design.

IMPACT AND TRANSFORMATION

The availability of electricity drastically increases the quality of healthcare provided. Refrigeration is critical to conserving valuable vaccines. Sterilization measures require energy. With community-generated power, locals no longer have to travel long distances to the city for health care treatment. Also, communities are well lighted hence improving overall security and safety.

Pilot project outcome also include improved educational services, increased economic activities, lower cost of powering businesses, an expanded women and youth empowerment. Darway will power light manufacturing such as a water bottling company, a variety of agricultural processing and packaging, and hospitality services in the communities served. Darway's plans include the powering of telecom base stations in the communities which will provide affordable internet access. . Darway aims to continuously engage and empower beneficiaries' communities through various post implementation value added services.

TIMELINE

This is a 15-month program. By Q3 2023, Darway Coast plans to develop and implement 11 isolated minigrids. The project will provide energy access to over 15,000 customers and improve the lives of over 80,000 people. The project will create over 200 permanent and temporary jobs while increasing local capacity.

AT A GLANCE

Project Locations:	Nigeria
GPS	5.034771, 6.676498, 5.961397, 7.435593
IEEE Funding	US$ 200,000.00
Payment Tranches	US$170,000/Q2 2022, US$20,000/Q4 2022 US$10,000/Q1 2023
# local IEEE members involved	STAFF MEMBERS AND VOLUNTEERS WHO ARE IEEE MEMBERS
Number of Beneficiaries (Phase 1)	20,000
Number of Beneficiaries (At scale)	80,000
Technologies to be deployed	Solar Hybrid Minigrids
IEEE Review and Reporting Period	Monthly & Quarterly
Sustainability/ Scalability	2 years
Project Contact	Henry Ureh, darway@darwaycoast.com
Organization information	1 Morenikeji Street, Off Allen Avenue, Ikeja, Lagos, Nigeria